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Two-dimensional bound excitons in the real space and Landau quantization space: a comparative study

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The Landau quantization space is based on the respective motion of the electron and hole in a magnetic field and can provide a new route to understand the bound exciton behaviors observed in the experiments. In this paper, we study the two-dimensional exciton properties of monolayer WSe_2 in both the real space and Landau quantization space. Focusing on the excitons of zero center-of-mass momentum, we calculate its energy spectrum in both spaces, with the results agreeing well with each other. We then obtain the diamagnetic coefficients and root-mean-square radius, which are consistent with the s state results from the experiments. More importantly, in the exciton state nl, we find that the dominant electron-hole pair component may shift with the magnetic field and the Coulomb interactions, and reveal that the magnetic field will drive the dominant component to be the free electron-hole pair $\{n_e = n + l - 1, n_h = n - 1\}$, whereas the Coulomb interactions drives it to be the pair of the lower index.

I. INTRODUCTION

Recently, the monolayer transition-metal dichalcogenides (TMDs) have attracted tremendous research interests due to their remarkable properties that show potential applications in optoelectronics and valleytronics [1, 2]. In monolayer TMDs, the broken inversion symmetry can lead to a direct band gap in the low-energy valleys at the corners of the hexagonal Brillouin zone [3]. As a result, the strong light-matter couplings are allowed and can induce the optical excitations of the exciton states that are made up of the pair of an electron and a hole [4]. In two-dimensional (2D) systems, as the dimension is reduced, the Coulomb interactions between the electrons and holes get enhanced [5–7]. Consequently, the excitons will be tightly bound and have a much larger binding energy than those in conventional 2D semiconductors, such as the single GaAs quantum well [8] and GaAs/AlGaAs quantum wells [9].

Experimentally, the excitons in monolayer TMDs can be identified through the salient excitonic resonances in the photoluminescence or reflectance spectroscopy conducted under magnetic fields [10–15]. To unambiguously resolve these exciton states, the high-quality samples are required, which have been markedly improved due to the fabrication technique developments in recent years. The identifications of the exciton states in the experiments are of certain significance, *e.g.*, they can provide valuable information about the enhanced Coulomb interactions between the electrons and holes and the interaction-driven novel phase in such 2D systems.

As the magnetic field can polarize the spin and valley degrees of freedom of resident carriers in monolayer TMDs, it has become a useful tool to explore the exciton properties in the experiments [14–17]. Under a magnetic field, a free electron or hole will be deflected by the Lorentz force to move on curved orbits with a set of discrete energies, or called the Landau levels (LLs); while a charge-neutral exciton is not subjected to the Lorentz force. In Ref. [18], the authors proposed a thought-provoking question: how can a bound exciton state under a magnetic field be expressed through the free electron and hole LL components. In many previous theoretical works [3, 19–21], although the real space was commonly used to study the exciton properties, it cannot solve the question of the exciton state composition. On the other hand, the Landau quantization space provides a new route to understand the exciton behaviors, whose basis is composed of the products of the free electron and hole LL wave functions. In the Landau quantization space, when the Coulomb potential energy is accurately formulated, the above composition question can be well addressed [18]. However, a full comparison of the exciton energy spectrum under a magnetic field in the real space and Landau quantization space is still lacking, which motivates the present work. Moreover, in an exciton state, it is interesting to explore whether the exciton state composition could be modulated by external factors, *e.g.*, the magnetic field and Coulomb interactions?

In this paper, we will make a comparative study of 2D exciton states in monolayer TMDs under a magnetic field in both the real space and Landau quantization space. We take monolayer WSe_2 as an example and focus on the excitons of zero center-of-mass momentum, $K = 0$, or called the excitons in the light cone. The calculation methods in the two spaces will be detailed at length in Secs. III(A) and III(B), respectively. The obtained main results are given as follows: (i) We solve the energy spectrum of the bound excitons in the two spaces and find that the results agree well with each other. We point out that the magnetic quantum number l can be endowed with the meaning of pairing the electrons and holes as $n_h = n_e + l$ [18], in which $n_{e(h)}$ denotes the electron (hole) LL index. (ii) In the exciton state nl, according to the zero-field wave function as well as the

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quadratic fittings to the energy, we obtain the diamagnetic coefficient σ_{nl} and root-mean-square (rms) radius r_{nl}. The results show good agreement with the s state results in the experiments [12]. (iii) More importantly, in the Landau quantization space, we reveal that in an exciton state, the dominant free electron-hole pair may shift with the magnetic field and Coulomb interactions. We further calculate the phase diagrams of the dominant pair component of the exciton states. The phase diagrams indicate that in an exciton state nl, the magnetic field will drive the dominant component to be the electron-hole pair $\{n_e = n + l - 1, n_h = n - 1\}$, whereas the Coulomb interactions drive it to be the pair of the lower index; their competitions would determine the exciton state composition. Our theoretical study could provide deeper insights into the exciton states in monolayer TMDs and pave the way for their applications in future electronic devices.

II. MODEL

Consider 2D bound excitons in monolayer WSe_2. Within the effective mass approximation, when the magnetic field is absent, the model Hamiltonian describing the exciton dynamics is written as [21]

$$H_0 = \frac{\boldsymbol{p}_e^2}{2m_e} + \frac{\boldsymbol{p}_h^2}{2m_h} + V(r), \tag{1}$$

where $\boldsymbol{p}_{e(h)}$ denote the momentum for the electron (hole) and $m_{e(h)}$ is the mass, $V(r)$ is the Coulomb interaction potential, and $r = |\boldsymbol{r}_e - \boldsymbol{r}_h|$ is the electron-hole separation. In 2D electron gas, the nonlocal screening effects will enhance the Coulomb interactions [5–7], which can be described by the Keldysh potential that is given as [19, 22]

$$V(r) = -\frac{e^2}{8\epsilon_0 r_0}\Big[H_0\Big(\frac{\epsilon_v r}{r_0}\Big) - Y_0\Big(\frac{\epsilon_v r}{r_0}\Big)\Big]. \tag{2}$$

Here $H_0(x)$ and $Y_0(x)$ are the Struve function and Bessel function of the second kind, respectively, ϵ_0 is the vacuum dielectric constant, $r_0 = 2\pi\chi_{2D}$ denotes the screening length characterized by the dielectric nature of monolayer WSe_2, with χ_{2D} being the 2D polarizability, and ϵ_v is the relative dielectric constant that is related to the encapsulating hexagonal BN slabs. For large electron-hole separation $r \gg r_0$, the Keldysh potential behaves as a Coulomb-like potential, $V(r) \sim \frac{1}{r}$; whereas for small separation $r \ll r_0$, it diverges weakly as $V(r) \sim \log(r)$. Thus, the Keldysh potential is expected to induce the exciton states that are markedly different from those described by the hydrogen-like potential.

When a uniform magnetic field $\boldsymbol{B} = B\hat{e}_z$ is applied on the 2D system, we use the Peierls substitution to replace the momenta as $\boldsymbol{p} \to \boldsymbol{p} - q\boldsymbol{A}$, in which $\boldsymbol{A}$ is the magnetic vector potential and $q = \mp e$ is the charge for the electron and hole, respectively. Then the Hamiltonian becomes

$$H = \frac{(\boldsymbol{p}_e + e\boldsymbol{A}_e)^2}{2m_e} + \frac{(\boldsymbol{p}_h - e\boldsymbol{A}_h)^2}{2m_h} + V(r). \tag{3}$$

We will solve H in both the real space and the Landau quantization space, with the calculation methods presented explicitly in Sec. III. In the following, we set the model parameters extracted from the experiments [12] as $r_0 = 5$ nm, $\epsilon_v = 3.97$, $m_e = 0.29m_0$, $m_h = 0.64m_0$, and m_0 is the electron mass.

III. CALCULATION METHODS

A. The real space

In the real space, we switch to the center-of-mass and relative motion coordinates, $\boldsymbol{R} = \frac{1}{M}(m_e\boldsymbol{r}_e + m_h\boldsymbol{r}_h)$ and $\boldsymbol{r} = \boldsymbol{r}_e - \boldsymbol{r}_h$, with $M = m_e + m_h$ being the total mass. For the Schrödinger equation $H\Psi = \varepsilon\Psi$, we perform the gauge transformation to the wavefunction $\Psi \to e^{if}\Psi$. When we choose the gauge $f = \frac{eB}{2\hbar}(x_e y_h - y_e x_h)$ [23], the Hamiltonian will be transformed as $H \to e^{-if}He^{if}$, which is given as

$$H = \frac{\big(\frac{m_e}{M}\boldsymbol{P} + \boldsymbol{p} + e\boldsymbol{A}_r\big)^2}{2m_e} + \frac{\big(\frac{m_h}{M}\boldsymbol{P} - \boldsymbol{p} + e\boldsymbol{A}_r\big)^2}{2m_h} + V(r). \tag{4}$$

Here $\boldsymbol{P}$ and $\boldsymbol{p}$ denote the center-of-mass momentum and relative momentum of the exciton, respectively. Focusing on the vanishing $\boldsymbol{P}$, we obtain the Hamiltonian for the relative motion of the electron-hole pair,

$$H = \frac{p^2}{2m_r} + \Big(\frac{1}{m_e} - \frac{1}{m_h}\Big)e\boldsymbol{p}\cdot\boldsymbol{A}_r + \frac{e^2A^2}{2m_r} + V(r), \tag{5}$$

where $m_r^{-1} = m_e^{-1} + m_h^{-1}$ is the reduced mass of the exciton. Adopting the symmetric gauge $\boldsymbol{A}_r = \frac{1}{2}B(\hat{\boldsymbol{e}}_z \times \boldsymbol{r})$, we have $\boldsymbol{p}\cdot\boldsymbol{A}_r = \frac{1}{2}BL_z$, with $L_z = -i\hbar\frac{\partial}{\partial\theta}$ denoting the angular momentum operator along the z direction. Thus, in Eq. (5), the first term gives the kinetic energy of the relative motion, the second term gives the interactions between the exciton magnetic moment and the magnetic field, which can be identified as the Zeeman shift, and the third term represents the weak quadratic confinement potential that determines the diamagnetic shift [23].

Since H owns the radial symmetry, we have the commutation relation $[H, L_z] = 0$. As a result, the wavefunction can be divided into the radial and angular components, $\Psi_{nl}(\boldsymbol{r}) = R_{nl}(r)Y_l(\theta)$, with $Y_l(\theta) = \frac{1}{\sqrt{2\pi}}e^{il\theta}$. Similar to the 2D hydrogen atom [24], we use the index nl to label the energy levels of the excitons, in which the principle quantum number $n = 1, 2, \cdots$, and the magnetic quantum number $l = 0, \pm1, \pm2, \pm3, \cdots, \pm(n-1)$, corresponding to the $s, p\pm, d\pm, f\pm, \cdots$ states. By inserting

Ψ_{nl} into Eq. (5), we obtain the reduced radial equation,

$$\Big[-\frac{\hbar^2}{2m_r}\Big(\partial_r^2+\frac{1}{r}\partial_r-\frac{l^2}{r^2}\Big)+\Big(\frac{1}{m_e}-\frac{1}{m_h}\Big)\frac{\hbar leB}{2}+\frac{e^2B^2}{8m_r}r^2 + V(r)\Big]R_{nl}(r)=\varepsilon_{nl}R_{nl}(r). \tag{6}$$

As Eq. (6) includes the Keldysh potential $V(r)$, it cannot be solved analytically and we resort to numerics [13]. To numerically solve the second-order differential equation, we introduce the first derivative of the radial wave function

$$D_{nl}(r)=\partial_r R_{nl}(r). \tag{7}$$

In the calculations, both $R(r)$ and $D(r)$ will be represented on the equidistant grid points r_i from the origin $r_1=0$ to the cutoff $r_N=r_{\rm cut}$. Here $r_{\rm cut}$ should be large enough to indicate the asymptotic behavior at infinity. On the other hand, the separation between neighboring grids $\delta r=r_{i+1}-r_i$ should be small enough to ensure the accuracy of the results. In the following, we will set $\delta r=0.05$ nm and $r_{\rm cut}=60$ nm (see Appendix).

We transform the differential equations (6) and (7) into finite-difference equations. Explicitly, Eq. (6) transforms into a set of linear equations,

$$-\frac{\hbar^2}{2m_r}\Big[\frac{D(r_{i+1})-D(r_i)}{r_{i+1}-r_i}+\frac{D(r_{i+1})+D(r_i)}{r_{i+1}+r_i}\Big]+\Big[\frac{\hbar^2l^2}{2m_r(\frac{r_i+r_{i+1}}{2})^2}+\frac{e^2B^2}{8m_r}\Big(\frac{r_{i+1}+r_i}{2}\Big)^2+\Big(\frac{1}{m_e}-\frac{1}{m_h}\Big)\frac{\hbar leB}{2} + V\Big(\frac{r_i+r_{i+1}}{2}\Big)\Big]\times\frac{R(r_{i+1})+R(r_i)}{2}=\varepsilon\times\frac{R(r_{i+1})+R(r_i)}{2}, \tag{8}$$

and Eq. (7) transforms into another set of linear equations,

$$\frac{1}{2}\big[D(r_{i+1})+D(r_i)\big]=\frac{R(r_{i+1})-R(r_i)}{r_{i+1}-r_i}. \tag{9}$$

Note that in both Eqs. (8) and (9), the index i runs from $i=1$ to $i=N-1$. To solve the $2N$ unknowns, we consider two boundary conditions: (i) the vanishing wavefunction at $r_N=r_{\rm cut}$ and (ii) the vanishing derivative at $r_1=0$, which are given as

$$R(r_N)=0,\quad D(r_1)=0. \tag{10}$$

The $2N-2$ equations in Eqs. (8) and (9) together with the two boundary condition equations in Eq. (10) constitute $2N$ linear equations with $2N$ variables. Since only $N-1$ equations in Eq. (8) include the eigenenergy ε, such $2N$ equations do not formulate a standard matrix eigenvalue problem, but instead a generalized eigenvalue problem,

$$AV=\varepsilon BV. \tag{11}$$

Here the matrix A incorporates the left-hand sides of Eqs. (8), (9) and (10), and V is a vector containing the total $2N$ components of $R(r_i)$ and $D(r_i)$ on the grid points. In matrix B, the elements are mostly zero except for those corresponding to the right-hand side of Eq. (8). In the calculations, we use the function dggev.f in the Lapack library to solve Eq. (11).

B. The Landau quantization space

Consider a free electron or hole in a 2D system with area $L_x\times L_y$. Under a perpendicular magnetic field $\boldsymbol{B}=B\hat{\boldsymbol{e}}_z$, we choose the vector potential in the Landau gauge as $\boldsymbol{A}=Bx\hat{\boldsymbol{e}}_y$. Then the Hamiltonian for the electron (hole) is written as

$$H_{e(h)}=\frac{1}{2m_{e(h)}}\big[p_{e(h)x}^2+(p_{e(h)y}\pm eBx)^2\big]. \tag{12}$$

The upper (lower) sign denotes the electron (hole) case. Since $[H_{e(h)},p_{e(h)y}]=0$, the wave vector $k_{e(h)y}$ remains as a good quantum number. The eigenenergy and eigenstate of the n-th LL are given as

$$\varepsilon_{n_{e(h)}}=\Big(n_{e(h)}+\frac{1}{2}\Big)\hbar\omega_{e(h)}, \tag{13}$$

$$\psi_{n_{e(h)},k_{e(h)y}}(x,y)=\frac{e^{ik_{e(h)y}y}}{\sqrt{L_yl_B}}\tilde{H}_{n_{e(h)}}\Big(\frac{x-x^*_{e(h)}}{l_B}\Big). \tag{14}$$

where $\omega_{e(h)}=\frac{eB}{m_{e(h)}}$ is the cyclotron frequency, $x^*_{e(h)}=\mp k_{e(h)y}l_B^2$ gives the equilibrium position of the oscillations, $l_B=\sqrt{\frac{\hbar}{eB}}$ is the magnetic length, $\tilde{H}_n(x)$ denotes the Hermite-Gaussian function that is made up of the product of the Hermite polynomial $H_n(x)$ and the Gaussian weight function

$$\tilde{H}_n(x)=\frac{1}{\sqrt{2^nn!\sqrt{\pi}}}e^{-x^2/2}H_n(x). \tag{15}$$

Note that in Eq. (13), as the LL energy is independent of $k_{e(h)y}$, the LLs are highly degenerate, with the degeneracy determined by $k_{e(h)y}$ as $N_L=\frac{L_xL_y}{2\pi l_B^2}$.

When transforming to the momentum space, the wave function becomes

$$\psi_{n_{e(h)}}(k_{e(h)x},k_{e(h)y}) = (-i)^{n_{e(h)}}\sqrt{\frac{2\pi l_B}{L_x}}e^{\pm ik_{e(h)x}k_{e(h)y}l_B^2} \times \tilde{H}_{n_{e(h)}}(k_{e(h)x}l_B), \tag{16}$$

where the phase factor $e^{\pm ik_{e(h)x}k_{e(h)y}l_B^2} = e^{-ik_{e(h)x}x^*_{e(h)}}$ comes from the equilibrium position $x^*_{e(h)}$. In the second-quantization form, we rewrite the wavefunction as [18]

$$|n_{e(h)},k_{e(h)y}\rangle = i^{-n_{e(h)}}\sqrt{\frac{2\pi l_B}{L_x}}\sum_{k_x}e^{\pm ik_{e(h)x}k_{e(h)y}l_B^2} \times \tilde{H}_n(k_{e(h)x}l_B)|\boldsymbol{k}_{e(h)}\rangle, \tag{17}$$

where $|\boldsymbol{k}_{e(h)}\rangle = c^\dagger_{\boldsymbol{k}_{e(h)}}|0\rangle$. Note that $|n_{e(h)},k_{e(h)y}\rangle$ constitutes the basis of the Landau quantization space.

We consider the direct interactions between the two particles 1 and 2. In the basis $|n_{e(h)},k_{e(h)y}\rangle$, the matrix element for the interaction potential is given as $V^{n_1',k_{y1}';n_2',k_{y2}'}_{n_1,k_{y1};n_2,k_{y2}} = \langle n_1',k_{y1}';n_2',k_{y2}'|\hat{V}|n_1,k_{y1};n_2,k_{y2}\rangle$. Here the index without and with prime denote the initial and final states, respectively. After a direct calculation, we obtain [18]

$$V^{n_1',k_{y1}';n_2',k_{y2}'}_{n_1,k_{y1};n_2,k_{y2}} = \sum_{\boldsymbol{q}}e^{\mp iq_xk_{y1}l_B^2\pm iq_xk_{y2}l_B^2}S^{n_1'}_{n_1}(\boldsymbol{q})S^{n_2'}_{n_2}(-\boldsymbol{q}) \times V(q)\delta_{k_{y1}',k_{y1}+q_y}\delta_{k_{y2}',k_{y2}-q_y}. \tag{18}$$

Similarly, the upper (lower) sign denotes the electron (hole) case. Here $V(q)$ is the interaction potential in the momentum space, the two δ-functions indicate that the momentum is conserved in the scattering process, and $S^{n'}_n(\boldsymbol{q})$ is the form factor given by

$$S^{n'}_n(\boldsymbol{q}) = i^{\Delta n}e^{\mp iq_xq_yl_B^2}\frac{2\pi l_B}{L_x}\sum_{k_x}e^{\mp ik_xq_yl_B^2} \times \tilde{H}_n(k_xl_B)\tilde{H}_{n'}(k_xl_B+q_xl_B), \tag{19}$$

with the index difference $\Delta n = n-n'$. When transforming the summation over k_x into integration and further completing the integration by using the generating function of the Hermite polynomials, $S^{n'}_n(\boldsymbol{q})$ is calculated as

$$S^{n'}_n(\boldsymbol{q}) = i^{\Delta n}\sqrt{\frac{n!n'!}{2^{n+n'}}}e^{\mp\frac{i}{2}\alpha\beta}e^{-\frac{\alpha^2+\beta^2}{4}}\sum_{m=0}^{\min(n,n')}\frac{2^m}{m!} \times \frac{(-\alpha+i\beta)^{n-m}}{(n-m)!}\frac{(\alpha+i\beta)^{n'-m}}{(n'-m)!}, \tag{20}$$

where the dimensionless parameters are $\alpha = q_xl_B$ and $\beta = q_yl_B$. Similar expressions in Eqs. (18)-(20) of the interactions between particles under a magnetic field can also be found in several previous literatures [25–28].

To simplify Eq. (20), we define the orthonormal function $\tilde{L}^{n'}_n(x)$ as

$$\tilde{L}^{n'}_n(x) = \sqrt{\frac{n!}{(n+n')!}x^{n'}e^{-x}}L^{n'}_n(x), \tag{21}$$

where $L^{n'}_n(x)$ is the generalized Laguerre polynomial [29]. Then $S^{n'}_n(\boldsymbol{q})$ is rewritten as

$$S^{n'}_n(\boldsymbol{q}) = i^{|\Delta n|}e^{\mp\frac{i}{2}q_xq_yl_B^2\mp i\Delta n\theta}\tilde{L}^{|\Delta n|}_{\tilde{n}}\Big(\frac{q^2l_B^2}{2}\Big), \tag{22}$$

where $\tilde{n}$=min$\{n,n'\}$ is the minimum between the initial state n and final state n'. In the numerical calculations, when compared to $L^{n'}_n(x)$, $\tilde{L}^{n'}_n(x)$ does not diverge even when the index n and n' are much large, and thus $\tilde{L}^{n'}_n(x)$ is more controllable and favorable.

Without the Coulomb interactions, the Schrödinger equation for a free electron-hole pair is

$$(H_e+H_h)|\Psi_0\rangle = (\varepsilon_{n_e}+\varepsilon_{n_h})|\Psi_0\rangle, \tag{23}$$

where the free electron-hole pair wave function is $|\Psi_0\rangle = |n_e,k_e;n_h,k_h\rangle = |n_e,k_e;n_h,K-k_e\rangle$, and $K = k_e+k_h$ is the center-of-mass wave vector along the y direction. Note that $k_{e(h)} = k_{e(h)y}$. When turning on the Coulomb interactions between the electrons and holes, we project the bound exciton states onto the space spanned by the free electron-hole pair states, which gives the kernel idea of solving the present interaction problem. This means that the bound exciton state can be expressed as a superposition of the free pair states,

$$|\Psi_K\rangle = \sum_{n_e,k_e;n_h}\phi_K(n_e,k_e;n_h)|\Psi_0\rangle, \tag{24}$$

where $\phi_K(n_e,k_e;n_h)$ is the expanding coefficient. Since the wave vector K is not changed by the Coulomb interaction, it is conserved and gives a constant of motion. It is interesting to see that as the separation between the equilibrium positions of the electron and hole is $x^*_h - x^*_e = Kl_B^2$, the separation also remains unchanged. Thus, the Schrödinger equation including the Coulomb interactions is written as

$$\big[H_e+H_h+V\big(|\boldsymbol{r}_e-\boldsymbol{r}_h|\big)\big]|\Psi_K\rangle = \varepsilon_K|\Psi_K\rangle, \tag{25}$$

from which the expanding coefficient ϕ_K can be solved.

We left-multiply Eq. (25) by the state $\langle\Psi_0'| = \langle n_e',k_e';n_h',K-k_e'|$ and arrive at

$$\sum_{n_e,n_h}\big[(\varepsilon_{n_e}+\varepsilon_{n_h})\delta_{n_e,n_e'}\delta_{n_h,n_h'}+\tilde{V}^{n_e',n_h'}_{n_e,n_h}\big]\phi_K(n_e,n_h) = \varepsilon_K\phi_K(n_e',n_h'), \tag{26}$$

where the contracted coefficient gives as $\phi_K(n_e,n_h) = \sum_{k_e}\phi_K(n_e,k_e;n_h)$, and the matrix element for the interaction potential $\tilde{V}^{n_e';n_h'}_{n_e;n_h} =$

$\sum_{\boldsymbol{q}} e^{-iKq_x l_B^2} V(q) S_{n_e}^{n_e'}(\boldsymbol{q}) S_{n_h}^{n_h'}(-\boldsymbol{q})$ is obtained as

$$\tilde{V}_{n_e,n_h}^{n_e',n_h'} = i^{|\Delta n_e|-|\Delta n_h|} \sum_{\boldsymbol{q}} e^{-iKq_x l_B^2} V(q) e^{-i(\Delta n_e - \Delta n_h)\theta} \times \tilde{L}_{\tilde{n}_e}^{|\Delta n_e|}\Big(\frac{q^2 l_B^2}{2}\Big) \tilde{L}_{\tilde{n}_h}^{|\Delta n_h|}\Big(\frac{q^2 l_B^2}{2}\Big). \tag{27}$$

Note that the matrix element $\tilde{V}$ in Eq. (27) is different from V in Eq. (18), as in the former, the δ-functions related to the wave vectors have been summed over in the calculations and thus will not appear. For the Keldysh potential $V(q)$ in the momentum space, it can be obtained through the Fourier transformation of $V(r)$, which is given as [31]

$$V(q) = -\frac{2\pi e^2}{q\epsilon(q)}, \tag{28}$$

with the effective dielectric constant $\epsilon(q) = 4\pi\epsilon_0(\epsilon_v + r_0 q)$.

We focus on the zero center-of-mass momentum exciton, $K = 0$. When turning the summation over the wave vector $\boldsymbol{q}$ into integration, the potential energy matrix element $\tilde{V}_{n_e,n_h}^{n_e',n_h'}$ becomes

$$\tilde{V}_{n_e,n_h}^{n_e',n_h'} = \frac{1}{4\pi^2} i^{|\Delta n_e|-|\Delta n_h|} \int_0^\infty q dq V(q) \times \tilde{L}_{\tilde{n}_e}^{|\Delta n_e|}\Big(\frac{q^2 l_B^2}{2}\Big) \tilde{L}_{\tilde{n}_h}^{|\Delta n_h|}\Big(\frac{q^2 l_B^2}{2}\Big) \int_0^{2\pi} d\theta e^{-i(\Delta n_e - \Delta n_h)\theta}. \tag{29}$$

After completing the integration over the angle θ, the nonvanishing matrix element $\tilde{V}$ requires the condition

$$\Delta n_e = \Delta n_h, \tag{30}$$

which gives the selection rules for the bound excitons [18]. Here, as the selection rules include both states before and after the scatterings, they are different from those in the LL transitions of the noninteracting Dirac electrons/holes [32–37].

If we define $l_k = n_e - n_h$ and $l_k' = n_e' - n_h'$, the above selection rules indicate that $l_k = l_k'$. As the potential energy matrix element vanishes for the scatterings between the electron-hole pairs of different l_k, i.e., $\tilde{V}_{i+l_k,i}^{i'+l_k',i'} = 0$ for $l_k \neq l_k'$, the Hamiltonian matrix can be partitioned into the block-diagonal form. Each block includes the electron-hole pairs $(n_e = i + l_k, n_h = i)$, or $(n_e = i, n_h = i + l_k)$ for $-l_k$. Finally, we arrive at the matrix eigenvalue equation for the bound exciton with the index l_k as

$$(\hat{T}_{l_k} + \hat{\tilde{V}}_{l_k})\Psi_{nl_k} = \varepsilon_{nl_k}\Psi_{nl_k}. \tag{31}$$

Here the kinetic energy and potential energy matrix elements are given as

$$T_{l_k}^{i,i'} = \Big[\hbar\omega_e\Big(i + l_k + \frac{1}{2}\Big) + \hbar\omega_h\Big(i + \frac{1}{2}\Big)\Big]\delta_{i,i'}, \tag{32}$$

$$\tilde{V}_{l_k}^{i,i'} = -\frac{e^2}{l_B}\int_0^\infty \frac{dx}{\epsilon(x)} \tilde{L}_{\tilde{i}+l_k}^{|\Delta i|}\Big(\frac{x^2}{2}\Big) \tilde{L}_{\tilde{i}}^{|\Delta i|}\Big(\frac{x^2}{2}\Big), \tag{33}$$

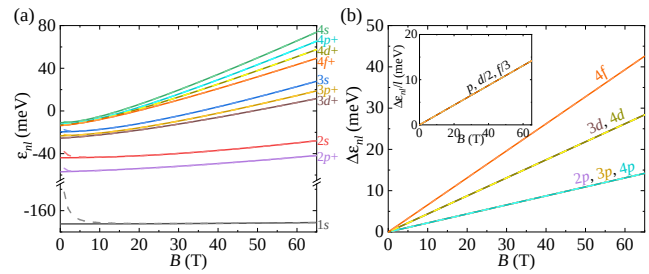

FIG. 1. (Color online) (a) The exciton energy spectrum ε_{nl} vs the magnetic field B. (b) The energy difference $\Delta\varepsilon_{nl} = \varepsilon_{n,+l} - \varepsilon_{n,-l}$ vs B. In both figures, the solid and dashed lines denote the results from the real space and Landau quantization space, respectively. The inset in (b) shows $\Delta\varepsilon_{nl}$ scaled by l, all of which collapse on the same line.

where $\Delta i = i' - i$, $\tilde{i} = \min\{i, i'\}$, and the dimensionless quantity $x = ql_B$, and the wave function $\Psi_{nl_k} = (\phi_{nl_k}^0, \phi_{nl_k}^1, \cdots)^T$, with the component $\phi_{nl_k}^i = \phi_{K=0}(i + l_k, i)$.

In the numerical calculations, completing the integration in Eq. (33) is more time-consuming than the matrix diagonalization in Eq. (31). This is because the function $\tilde{L}_i^{i'}(x)$ oscillates quickly with x when the index i and i' increase. Moreover, as the Hilbert space spanned by the electron-hole pairs is infinite, we need to truncate the matrix in Eq. (31) and set the dimension cutoff as $N_{\rm cut} = 300$ (see Appendix).

IV. MAIN RESULTS

A. Energy spectrum

First of all, we calculate the exciton energy spectrum ε_{nl} of monolayer WSe_2 in the real space and Landau quantization space, and plot ε_{nl} as a function of the magnetic field B in Fig. 1. Here, the results obtained from the two spaces are represented by the solid and dashed lines, respectively. In Fig. 1(a), when comparing ε_{nl} from the two spaces, we find that for the ground state $1s$, the energy ε_{1s} in the Landau quantization space exhibits certain deviations from that in the real space when $B < 12$ T, and becomes equivalent when $B > 12$ T. For the $2s$ state, ε_{2s} becomes equivalent when $B > 6$ T; the same case is found for the $2p+$ state. For the higher exciton states nl, ε_{nl} shows good agreement even under a weaker magnetic field. Such deviations of the lower energy levels under weak magnetic fields in the Landau quantization space can be attributed to the finite matrix dimension truncation that is associated with the poor convergence of the potential energy matrix element. For the higher exciton states and stronger magnetic field, the deviations will quickly disappear (see Appendix).

In Fig. 1(a), when $B = 0$, the energies of the ns states from the real space are given as $\varepsilon_{1s} = -172.44$ meV,

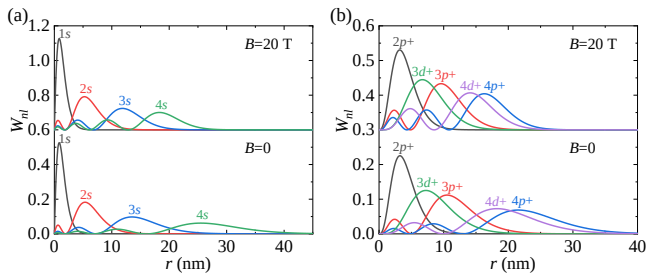

FIG. 2. (Color online) The probability W_{nl} of finding the bound exciton vs the electron-hole separation r, with $l = 0$ in (a) and $l = 1, 2$ in (b). In each lower and upper figure, the magnetic field is set as $B = 0$ and $B = 20$ T, respectively. For clarity, the probability curves under $B = 20$ T are shifted vertically by 0.6 in (a) and 0.3 in (b).

$\varepsilon_{2s} = -43.82$ meV, $\varepsilon_{3s} = -19.52$ meV, and $\varepsilon_{4s} = -10.96$ meV. Due to the Keldysh potential, these energy levels exhibit a markedly non-hydrogenic Rydberg series, and cannot be described within a hydrogen-like model. More importantly, these energy levels are consistent with the previous experimental and theoretical results on the exciton Rydberg states in monolayer WSe_2 [12, 14, 20], demonstrating the validity of our calculations. For the magnetic field over a large range $B \sim (0, 65$ T), the energy levels are grouped for the same n. With increasing B, the ground state $1s$ shows minor blueshift; whereas the excited states exhibit evident blueshifts, which become more apparent for the higher excited states. The blueshifts of the exciton states under magnetic fields are in good agreement with the experiments [10, 12, 14].

Consider the energy difference between the exciton states of positive and negative $l(l_k)$, $\Delta\varepsilon_{nl(l_k)} = \varepsilon_{n,+l(l_k)} - \varepsilon_{n,-l(l_k)}$. The results of $\Delta\varepsilon_{nl(l_k)}$ are displayed in Fig. 1(b), with $l(l_k)$ growing from $l(l_k) = 1$ to $l(l_k) = 3$. We observe that the lines of $\Delta\varepsilon_{nl}$ pass through the origin and overlap completely with those of $\Delta\varepsilon_{nl_k}$, indicating that $\Delta\varepsilon_{nl} = \Delta\varepsilon_{nl_k}$.

In the real space, the energy levels are solved from Eq. (6). When l changes to $-l$, we see that only the second term of the left-hand side will change and the other terms remain unchanged. As a result, the energy difference is

$$\Delta\varepsilon_{nl}(B) = \hbar(\omega_e - \omega_h)l. \tag{34}$$

Note that when l changes to $-l$ and B remains unchanged, the energy levels are equal to those when B changes to $-B$ and l remains unchanged, $\varepsilon_{n,-l}(B) = \varepsilon_{n,l}(-B)$. Thus, the energy difference is $\Delta\varepsilon_{nl} = 2l\mu_B^{\rm ex}B$, which is twice the Zeeman shift and originates from the orbital magnetic moment. The corresponding Bohr magneton is given as $\mu_B^{\rm ex} = \frac{e\hbar}{2}(\frac{1}{m_e} - \frac{1}{m_h})$. Clearly, $\mu_B^{\rm ex}$ is caused by unequal electron and hole masses and vanishes when $m_e = m_h$.

On the other hand, in the Landau quantization space, the state $|n, -l_k\rangle$ is obtained from the state $|n, l_k\rangle$ by making the electron-hole transformation $n_e \leftrightarrow n_h$. Under this transformation, the potential energy matrix element $\tilde{V}_{l_k}^{i,i'}$ in Eq. (33) remains unchanged, but the kinetic energy matrix element $T_{l_k}^{i,i'}$ in Eq. (32) will change, leading to the energy difference

$$\begin{aligned}\Delta\varepsilon_{nl_k} =& \hbar\Big[\omega_e\Big(n + l_k + \frac{1}{2}\Big) + \omega_h\Big(n + \frac{1}{2}\Big)\Big] \\ &- \hbar\Big[\omega_e\Big(n + \frac{1}{2}\Big) + \omega_h\Big(n + l_k + \frac{1}{2}\Big)\Big] \\ =& \hbar(\omega_e - \omega_h)l_k.\end{aligned} \tag{35}$$

Combining the observation $\Delta\varepsilon_{nl} = \Delta\varepsilon_{nl_k}$ with Eqs. (34) and (35), we have $l = l_k$. This means that although l_k originates from the selection rules of the electron-hole LL pairing in the Landau quantization space, $n_e = n_h + l_k$, and l denotes the magnetic quantum number in the real space, they are equivalent to each other in labeling the energy levels [18]. For simplicity, we drop the subindex k in the following. The above analysis of $\Delta\varepsilon_{nl}$ also indicates that the energy deviations of the lower levels under weak magnetic field in the Landau quantization space are not caused by the kinetic energy of the electron and hole cyclotron motion, but by the interaction potential energy.

Moreover, in Fig. 1(b), we observe that $\Delta\varepsilon_{nl}$ of the same l but different n lie on the same line. When scaling $\Delta\varepsilon_{nl}$ by l, they all coincide with $\Delta\varepsilon_{n's}$, as seen in Fig. 1(b) inset. These observations can be easily understood from Eqs. (34) or (35) that $\Delta\varepsilon_{nl}$ does not depend on n, but on l as $\Delta\varepsilon_{nl} = l\Delta\varepsilon_{n's}$.

Besides the energy spectrum, the real-space wavefunction $\Psi_{nl}(\boldsymbol{r}) = R_{nl}(r)Y_l(\theta)$ can also be obtained. According to $\Psi_{nl}(\boldsymbol{r})$, the probability $W_{nl}(r)dr$ of finding the bound exciton with the electron-hole separation in the range $(r, r + dr)$ is written as

$$W_{nl}(r)dr = \int_0^{2\pi} d\theta dr r R_{nl}^2(r) Y_l^2(\theta) = R_{nl}^2(r) r dr. \tag{36}$$

The results of W_{nl} are displayed in Fig. 2 as a function of r. In Figs. 2(a) and 2(b) when $B = 0$, we see that (i) for the ground state, W_{1s} has no node, while for the excited states, W_{nl} owns $n-l-1$ nodes (except $r = 0$ and $r = \infty$); (ii) for the state nl, W_{nl} will extend to larger r when n increases (or l decreases), meaning that the excitons are loosely bound. These results are the same as those in 2D hydrogen atoms [24]. At a finite magnetic field $B = 20$ T, the above conclusions do not change, but W_{nl} will be compressed to smaller r, suggesting that the excitons become more tightly bound.

B. Diamagnetic shift

In this section, we study the diamagnetic shift of the bound excitons. The diamagnetic shift is expected to increase quadratically with the magnetic field [23, 38, 39]

TABLE I. The diamagnetic coefficient σ_{nl} and rms radius r_{nl}, which are in units of μeV/T^2 and nm, respectively. The calculation results obtained from Eq. (38) as well as the fitting results in Fig. 3(b) are listed. For comparison, the experimental data in Ref. [12] are also listed.

exciton state	calculation results through Eq. (38)		fitting results in Fig. 3(b)		experimental data in Ref. [12]	
nl	σ_{nl}	r_{nl}	σ_{nl}	r_{nl}	σ_{nl}	r_{nl}
$1s$	0.309	1.68	0.306	1.67	0.31 ± 0.02	1.7 ± 0.1
$2s$	4.87	6.66	3.98	6.02	4.6 ± 0.2	6.6
$3s$	24.2	14.85	12.56	10.69	22 ± 2	14.3 ± 1.5
$4s$	76.2	36.34	23.31	14.56	/	/
$2p$	2.3	4.58	2.03	4.3	/	/
$3p$	15.7	11.96	9.19	9.14	/	/
$4p$	56.2	22.63	19.35	13.27	/	/
$3d$	9.43	9.27	5.94	7.35	/	/
$4d$	42.4	19.65	15.46	11.86	/	/
$4f$	28.4	16.08	11.29	10.13	/	/

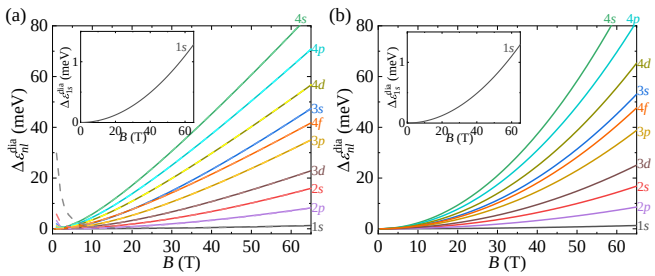

FIG. 3. (Color online) The diamagnetic shift $\Delta\varepsilon_{nl}^{\rm dia}$ vs the magnetic field B up to 65 T. In (a), $\Delta\varepsilon_{nl}^{\rm dia}$ is obtained from Eq. (38). The solid and dashed lines denote the results from the real space and Landau quantization space, respectively. In (b), $\Delta\varepsilon_{nl}^{\rm dia}$ is obtained by fitting the results in (a) through $\Delta\varepsilon_{nl}^{\rm dia} = \sigma_{nl}B^2$, from which the diamagnetic coefficient σ_{nl} and rms radius r_{nl} can also be solved and are listed in Table I. The insets in (a) and (b) show the results of the ground state $1s$.

and is taken as a measure of the confinement effect [40]. For the state nl, the diamagnetic shift is expressed as [38, 39]

$$\Delta\varepsilon_{nl}^{\rm dia}(B) = \sigma_{nl}B^2, \tag{37}$$

where σ_{nl} denotes the diamagnetic coefficient. The diamagnetic coefficient can determine the exciton effective mass, radius and the dielectric properties of a material [10, 11].

In Eq. (6), since only the second term of the left-hand side exhibits the quadratic field dependence, we obtain the diamagnetic shift as

$$\Delta\varepsilon_{nl}^{\rm dia}(B) = \frac{e^2}{8m_r}\langle r_{nl}^2\rangle B^2, \tag{38}$$

where the expectation value $\langle r_{nl}^2\rangle$ is with respect to the zero-field eigenstate [23]. The rms radius is related to σ_{nl} as $r_{nl} = \sqrt{\langle r_{nl}^2\rangle} = \frac{\sqrt{8m_r\sigma_{nl}}}{e}$ and can be used to estimate the spatial extent of the exciton wavefunctions. By using the zero-field wavefunction $\Psi_{nl}(\boldsymbol{r})$, we calculate $\langle r_{nl}^2\rangle$ as

$$\langle r_{nl}^2\rangle = \int_0^\infty rdr\int_0^{2\pi} d\theta r^2 R_{nl}^2(r)Y_l^2(\theta) = \int_0^\infty drr^3R_{nl}^2(r). \tag{39}$$

From $\langle r_{nl}^2\rangle$, the diamagnetic coefficient σ_{nl} and rms radius r_{nl} can be calculated, with the results listed in Table I. We see that both σ_{nl} and r_{nl} increase for the higher exciton state nl, reflecting the stronger diamagnetic behavior and larger rms radius. For comparison, in Table I, we also list the experimental data for the $1s$, $2s$ and $3s$ exciton states in monolayer WSe_2 [12]. We find that the theoretical results agree well with the experimental results, demonstrating the validity of the exciton model as well as the chosen parameters. In addition, we note that σ_{1s} and r_{1s} are close to those reported in monolayer WS_2 experiments [11].

Another route to calculate the diamagnetic shift is from the average energy of the nl and $n,-l$ states subtracting the zero-field energy [18],

$$\Delta\varepsilon_{nl}^{\rm dia}(B) = \frac{1}{2}\left[\varepsilon_{nl}(B) + \varepsilon_{n,-l}(B)\right] - \varepsilon_{n,l}(B=0). \tag{40}$$

Here the average energy cancels out the interactions between the exciton magnetic moment and the magnetic field, while the zero-field energy subtracts the contributions from the kinetic energy of relative motion and the Keldysh potential energy, as seen in Eq. (6). Following this way, we calculate $\Delta\varepsilon_{nl}^{\rm dia}$ and plot the results in Fig. 3(a), where the solid and dashed lines denote the results in the real space and Landau quantization space, respectively. Similar to Fig. 1(a), the diamagnetic shift obtained from the real space and Landau quantization space calculations agree with each other for the higher exciton states and stronger magnetic fields, but exhibit certain deviations for the lower exciton states and weaker magnetic fields. We fit the curves in Fig. 3(a) by using Eq. (37) and display the fitted results in Fig. 3(b). According to the fittings, we also obtain σ_{nl} and r_{nl}, which are listed in Table I. To see the results of the ground state

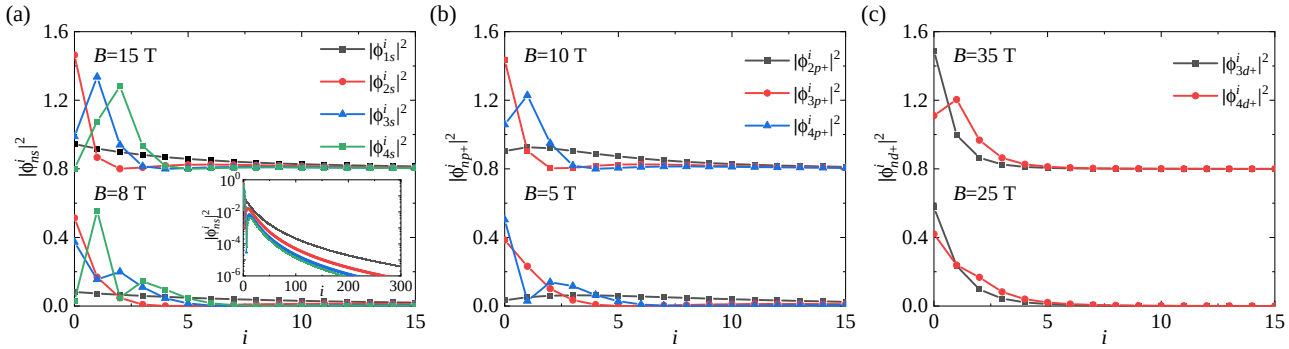

FIG. 4. (Color online) The weight $|\phi^i_{nl}|^2$ of the free electron-hole pair $\{n_e = i + l, n_h = i\}$ in the exciton state nl vs the hole LL index i. In (a)-(c), the weights are shown for the s, $p+$ and $d+$ states, respectively. The lower and upper figures are the results under different magnetic fields. For clarity, the weights in the upper figures are shifted vertically by 0.8. The inset in (a) shows the weight $|\phi^i_{ns}|^2$ vs the index i up to $i = 300$.

$1s$ more clearly, we show the calculated and fitted $\Delta\varepsilon^{\rm dia}_{1s}$ in the insets of Figs. 3(a) and (b), respectively.

Comparing the results from the zero-field wave function and the quadratic fittings, we find that for the three lowest exciton states, $1s$, $2s$ and $2p$, the results are consistent with each other; but for the other states, they exhibit certain differences and for the higher states, the difference is more apparent. For example, for the $4s$ state, σ_{nl} obtained from the zero-field wave function is more than three times that from the fittings. Such differences can be attributed to the fact that the diamagnetic shift represents a low-field behavior [23]; under a strong magnetic field, $\Delta\varepsilon^{\rm dia}_{nl}$ gradually changes from the parabolic dependence $\Delta\varepsilon_{nl} \sim B^2$ to the linear dependence $\Delta\varepsilon_{nl} \sim B$ [12], which will become more evident for the higher exciton states.

C. Exciton state composition

In the Landau quantization space, we can analyze the exciton state composition. That is, from the wave function Ψ_{nl}, the weight $|\phi^i_{nl}|^2$ of the free electron-hole pair $\{n_e = i + l, n_h = i\}$ can be extracted [18]. The composition of different exciton states are displayed in Fig. 4 as a function of the hole LL index i. Since the exciton states are mainly composed of the low-energy pairs, the weights are plotted from $i = 0$ to $i = 15$.

Intuitively, without the Coulomb interactions, the exciton state nl includes only the free electron-hole pair $\{n_e = n + l - 1, n_h = n - 1\}$; the Coulomb interactions can drive more electron-hole pairs into the exciton state. In Fig. 4(a), when the magnetic field $B = 8$ T and $B = 15$ T, the largest component of both the $1s$ and $2s$ states lies at $i = 0$, indicating that the $\{n_e = n_h = 0\}$ pair dominates the two states. This means that the Coulomb interactions do not change the dominant component of the $1s$ state, but will shift that of the $2s$ state from $i = 1$ to $i = 0$. In Fig. 4(a) inset, the weight $|\phi^i_{ns}|^2$ for the ns state is shown with the index i up to the cutoff $i = 300$. We see that when $i = 300$, the weight $|\phi^{300}_{1s}|^2$ of the ground state $1s$ reaches 10^{-5}; while the weights $|\phi^{300}_{2-4,s}|^2$ of other exciton states are vanishing. As the nonvanishing $|\phi^i_{1s}|^2$ can be extended to much larger i, the ground state includes more electron-hole pairs than other states, which can explain why in the Landau quantization space calculations, the convergence of the ground state energy is poor under weak magnetic fields (see Appendix).

Moreover, in Fig. 4(a) of the $3s$ ($4s$) state, the dominant component is $i = 0$ (1) when $B = 8$ T, and will shift to $i = 1$ (2) when the magnetic field increases to $B = 15$ T. Similar dominant component shift can also be found in other exciton states. In Fig. 4(b) of the $4p$ state, the dominant component is $i = 0$ when $B = 5$ T, and will shift to $i = 1$ when $B = 10$ T; in Fig. 2(c) for the $4d$ state, the dominant component is $i = 0$ when $B = 25$ T, and will shift to $i = 1$ when $B = 35$ T. These results indicate that with increasing B, the dominant component shift may occur in the exciton states.

To further explore the physics underlying the exciton state composition, we take the s states as an example and investigate how their dominant electron-hole pair component $\{n_e = n_h = i\}$ will be affected by the magnetic field B and Coulomb interactions. Here, we use the relative dielectric constant ϵ_v to control the interaction strength [41]. Experimentally, the dielectric environment can be effectively modulated by encapsulating the 2D flakes with different materials [10]. In the parametric space (B, ϵ_v), we calculate the phase diagram of the dominant electron-hole pair. The results of the $2s$, $3s$ and $4s$ states are displayed in Figs. 5(a)-5(c), respectively. In the ns state, as the dominant component can shift from the lowest pair of $i = 0$ up to the highest pair of $i = n - 1$, the total n different dominant components are included in each phase diagram. More observations are given as follows.

(i) When $\epsilon_v = 0$, the interaction strength reaches its maximum. For the field range $B \sim (0, 65$ T), in Fig. 5(a)

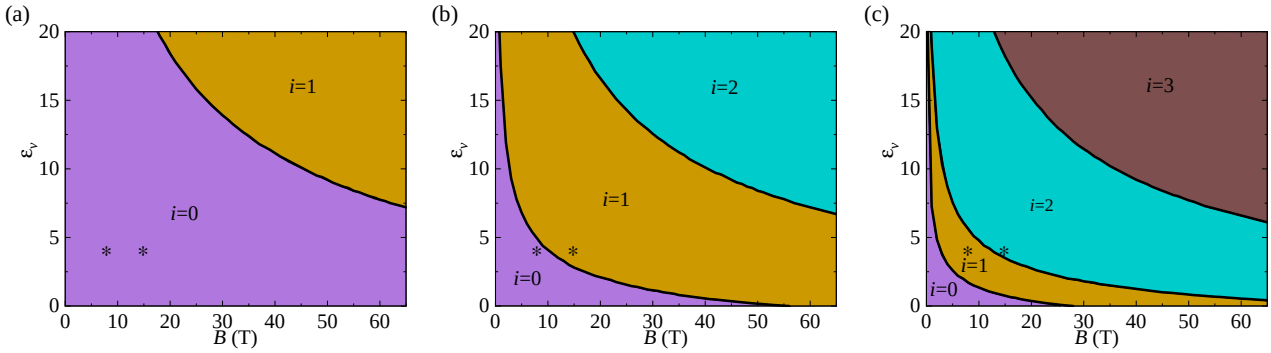

FIG. 5. (Color online) The phase diagrams of the dominant electron-hole pair $\{n_e = n_h = i\}$ in the parameter space spanned by the magnetic field B and the relative dielectric constant ϵ_v. (a)-(c) show the results of the $2s$, $3s$ and $4s$ states, respectively. The different dominant components i are labeled in different colors. The two asterisks marked in each figure correspond to the parameters chosen in Fig. 4(a).

of the $2s$ state, the dominant component remains as $i = 0$. By comparison, in Fig. 5(b) of the $3s$ state and Fig. 5(c) of the $4s$ state, the dominant component will both shift from $i = 0$ to $i = 1$ when B crosses a critical field.

(ii) With increasing ϵ_v, the interaction strength decreases. The magnetic field can drive the dominant component to shift from $i = 0$ to $i = n - 1$ successively, in which the critical fields separating the different components gradually decrease. Note that the two asterisks marked in each figure correspond to the two sets of parameters (B, ϵ_v) chosen in Fig. 4(a), from which the dominant component shift in the $3s$ and $4s$ states can be understood.

(iii) In the limit $\epsilon_v \to \infty$, the interaction strength tends to be zero. Although such a limiting case is not seen in the phase diagram, the asymptotic behavior that the component $i = n - 1$ will dominate the ns state can still be expected. This is consistent with the above intuitive analysis of the case without the Coulomb interactions.

Based on these observations, we reveal that in the ns states, the magnetic field will drive the dominant component to be the electron-hole pair of $i = n - 1$, while the Coulomb interactions will drive the dominant component to the pair of the lower i that owns the lower energy; their competitions determine the dominant component behaviors in Fig. 5. We suggest that the roles played by the magnetic field and Coulomb interactions in determining the dominant component can also be found in other exciton states, p, d, f, $\cdots$.

V. DISCUSSIONS AND SUMMARIES

To summarize, in this paper, we make a comparative study of the 2D bound excitons in monolayer WSe_2 under a magnetic field in both the real space and Landau quantization space. We obtain the energy spectrum and the diamagnetic shift, with the results in the two spaces agreeing well with each other. Although the two spaces have different physical pictures, in which the real space focuses on the relative motion of the electron and hole, while the Landau quantization space is based on respective cyclotron motion of the electron and hole in a magnetic field, they complement each other and can provide mutual validation supportive perspectives in understanding the exciton behaviors.

In the Landau quantization space, we analyze the exciton state composition and reveal the roles played by the magnetic field and Coulomb interactions. Since in most cases, the dominant component of an exciton state can have the weight of surpassing 50%, it directly determines some properties of the excitons, *e.g.*, the optical absorption and emission. We believe that the Landau quantization space can also be used to analyze the exciton state composition in other monolayer TMDs [11, 30, 31, 41], MoS_2, $MoSe_2$, and WS_2, as well as in TMD heterojunction moiré superlattices [42–44].

VI. ACKNOWLEDGMENTS

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VII. APPENDIX

Here we provide more details for the convergence of our numerical calculations.

Figure A1(a) shows the three lowest energy levels of the bound excitons obtained in the real space, which are given as a function of the magnetic field B for a set of the grid separation and the cutoff grid size $(\delta r, r_{\rm cut})$. We see that ε_{1s} converges quickly with the smaller δr and larger $r_{\rm cut}$; while ε_{2s} and ε_{2p+} show negligible variation for different $(\delta r, r_{\rm cut})$. These results indicate that in the real space, the convergence of the energy levels can be

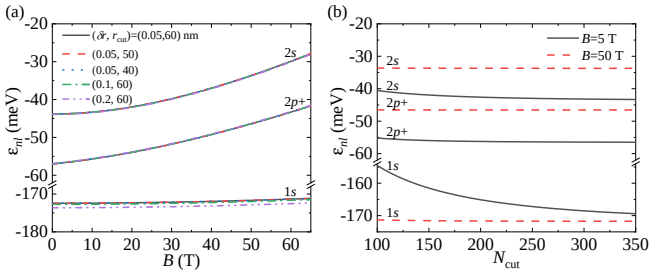

FIG. A1. (Color online) The convergence of the three lowest energy levels of the bound excitons. (a) The energy levels obtained from the real space calculations vs the magnetic field B for a set of the grid separation and the cutoff grid size $(\delta r, r_{\rm cut})$. (b) The energy levels obtained from the Landau quantization space calculations vs the matrix dimension cutoff $N_{\rm cut}$ when $B = 5$ T and 50 T.

quickly reached. To make the results more reliable, we choose $(\delta_r, r_{\rm cut}) = (0.05, 60)$ nm in the main text.

Figure A1(b) shows the three lowest energy levels obtained in the Landau quantization space, which are given as a function of the matrix dimension cutoff $N_{\rm cut}$ when $B = 5$ T and 50 T. At the weak magnetic field $B = 5$ T, we see that ε_{1s} decreases slowly with $N_{\rm cut}$ and will remain unchanged when $N_{\rm cut}$ is large enough, meaning that it converges to the accurate result. For ε_{2s} and ε_{2p+}, the convergence is reached at a smaller $N_{\rm cut}$. By comparison, at the strong magnetic field $B = 50$ T, we observe that the energy levels are almost unchanged with $N_{\rm cut}$. According to these results, we suggest that in the Landau quantization space, the convergence is poor for the lower energy levels and weak magnetic field, which can be attributed to the finite matrix dimension truncation that is associated with the potential energy matrix element, but can be quickly reached for the higher energy levels and the strong magnetic field. Considering our computational resources, we set $N_{\rm cut} = 300$ in the main text.

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